Rest Apis

What is Rest Apis?

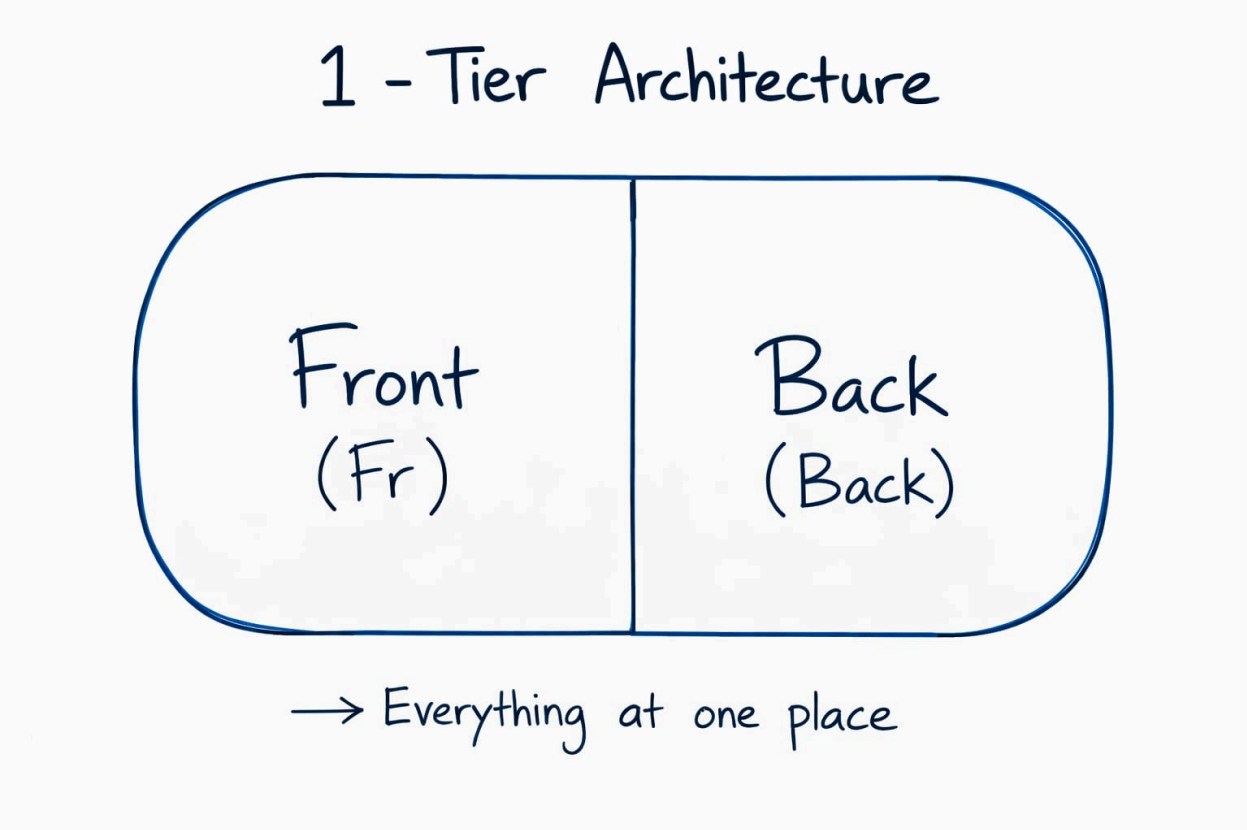
A **REST API** (Representational State Transfer Application Programming Interface) is an architectural style for networked applications that utilizes standard web protocols, primarily **HTTP**. It enables communication between clients and servers, allowing for the exchange of data and performance of actions in a standardized, stateless manner

Benefits of REST APIs

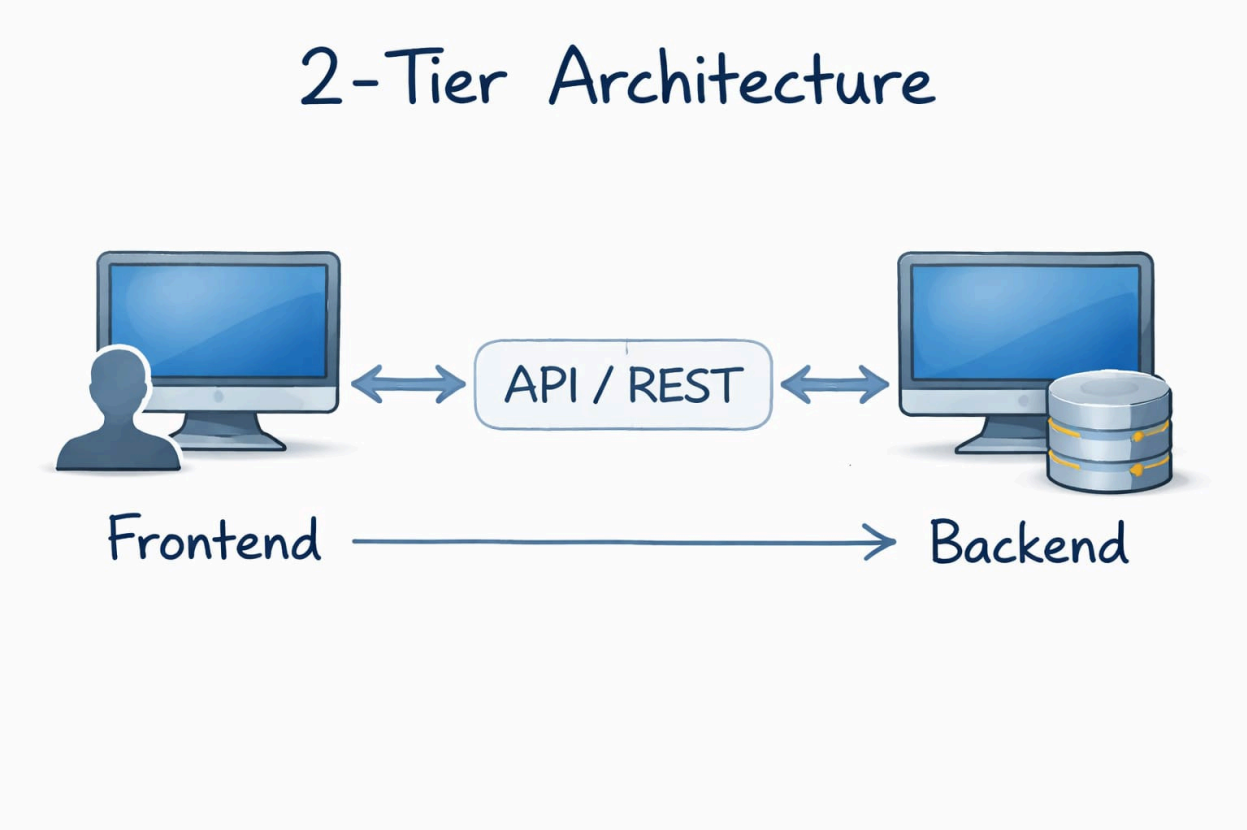
- **Scalable** – Stateless requests make it easier to scale applications.
- **Platform Independent** – Works with any client (web, mobile, desktop).
- **Simple to Use** – Uses standard HTTP methods like GET, POST, PUT, DELETE.
- **Flexible Data Format** – Usually uses JSON, which is lightweight and easy to parse.
- **Separation of Concerns** – Frontend and backend can be developed independently.
- **Easy Integration** – Can easily connect with third-party services and systems.

Architecture

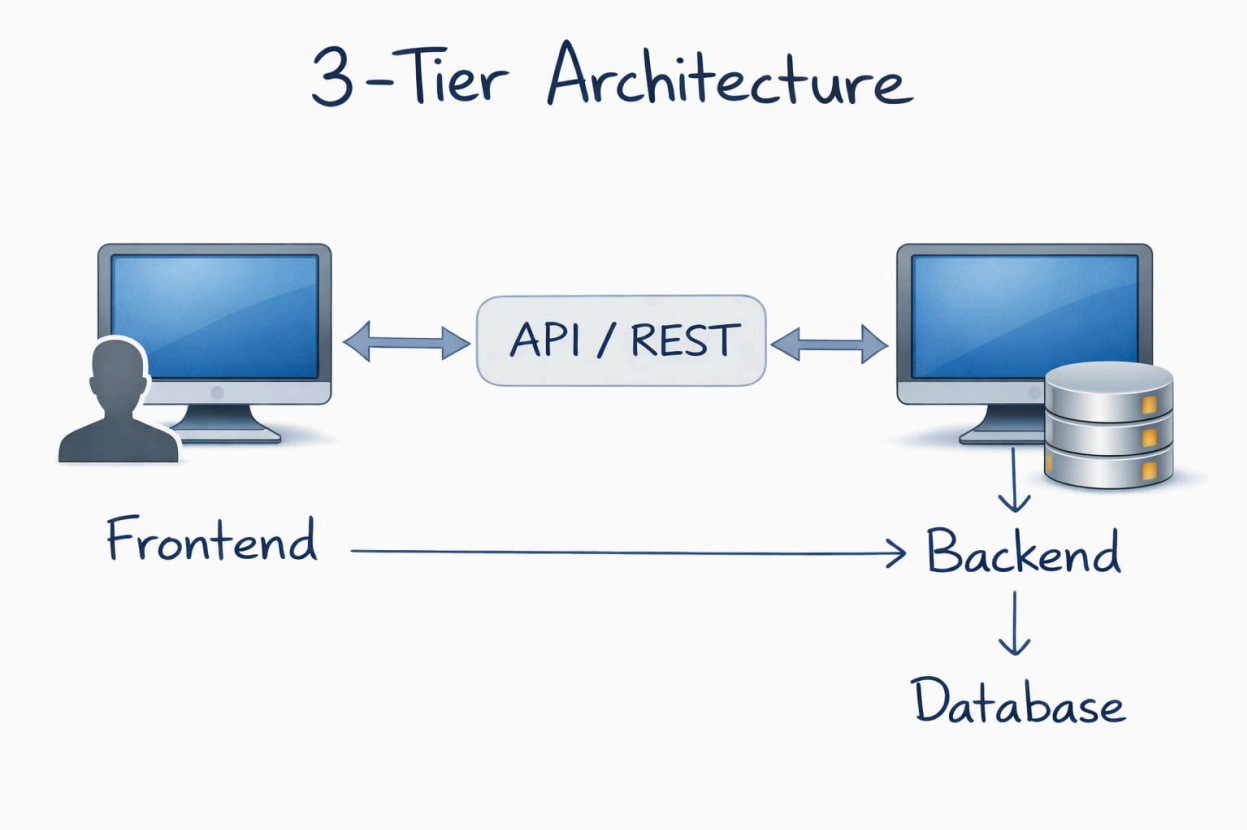
1 tier


1 - Tier Architecture
Front
(Fr)
Back
(Back)
→ Everything at one place

2 tier

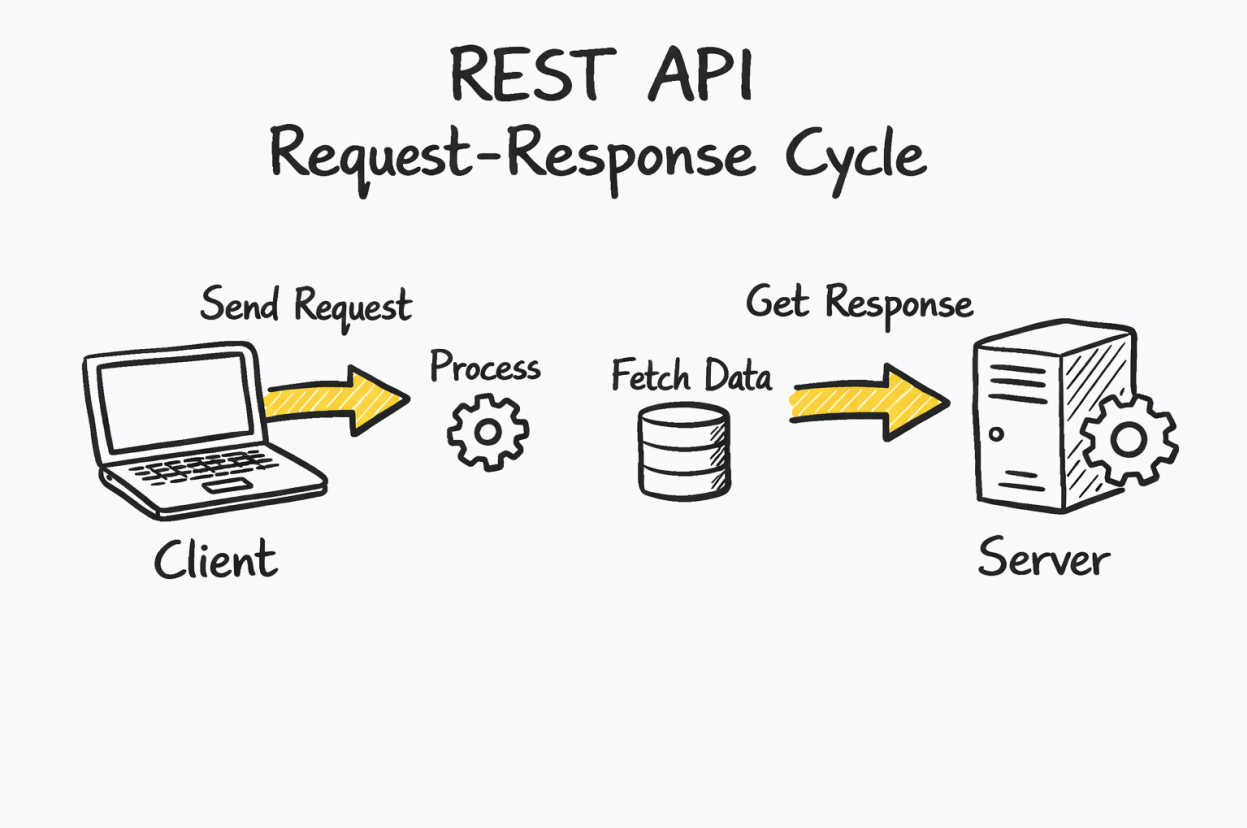


3 tier



Building Blocks

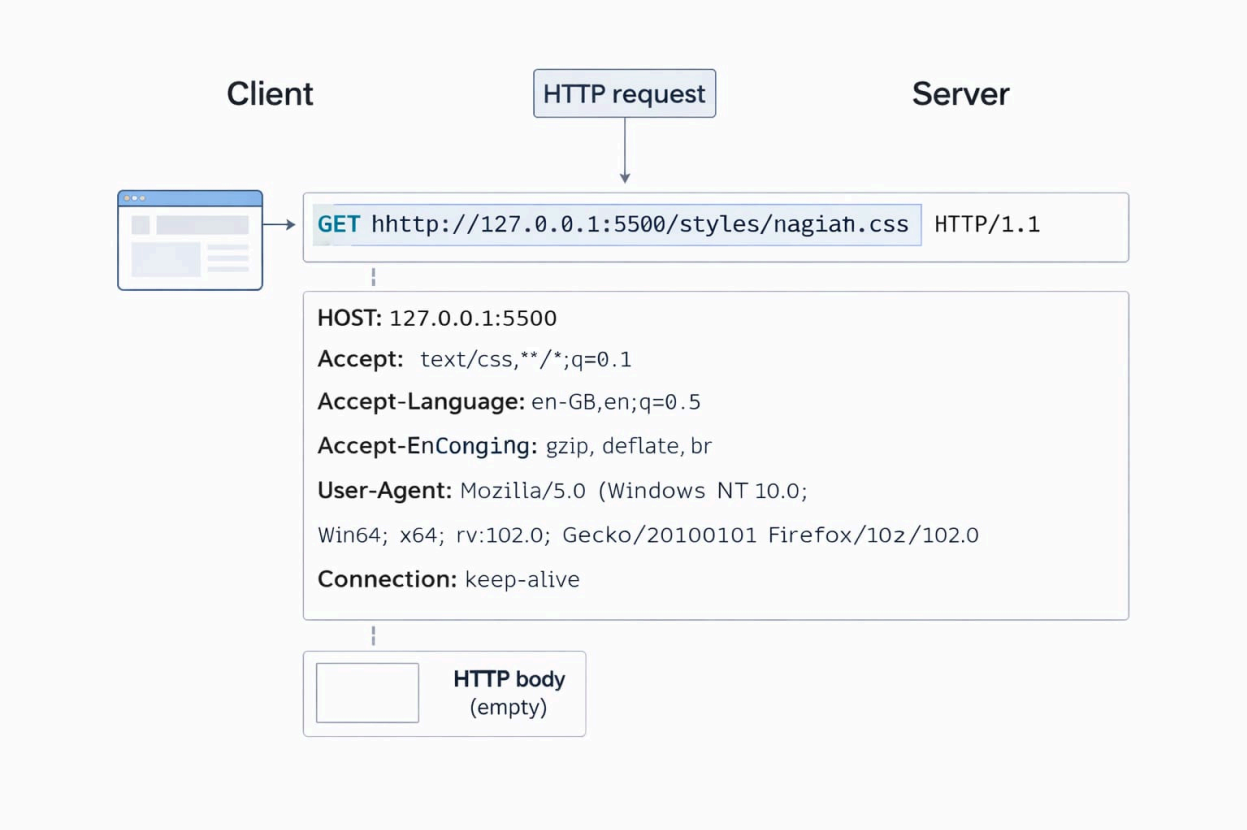
Request - Response



In the **request–response cycle**, a **client sends an HTTP request** to a server asking for data or an action to be performed.

The **server processes the request and sends back an HTTP response**, usually containing data (like JSON) or a status message.

Request



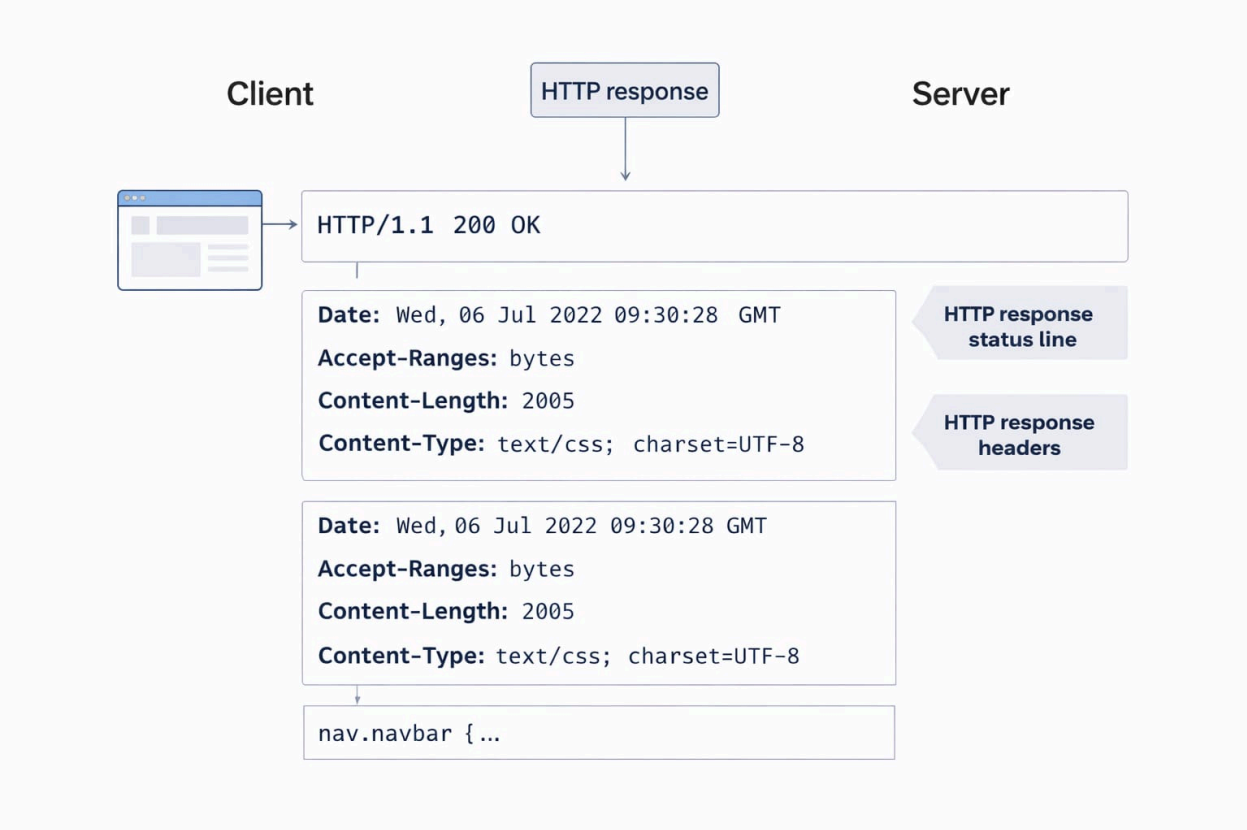
HTTP Request

An **HTTP request** is sent by a **client (browser)** to a **server** to request a resource such as a webpage, CSS file, or API data.

It mainly contains three parts:

- **Request Line** – Includes the **HTTP method**, **URL**, and **HTTP version** (e.g., `GET /styles.css HTTP/1.1`).
- **Request Headers** – Provide metadata about the request like **Host, User-Agent, Accept, Accept-Language, Connection**, etc.
- **Request Body** – Optional data sent with the request (usually used in **POST, PUT, PATCH** requests). For **GET requests**, the body is typically empty.

Response



HTTP Response

An **HTTP response** is sent by the **server back to the client** after processing an HTTP request.

It mainly contains three parts:

- **Status Line** – Includes the **HTTP version**, **status code**, and **status message** (e.g., `HTTP/1.1 200 OK`), which indicates whether the request was successful or failed.
- **Response Headers** – Provide metadata about the response such as **Date, Content-Type, Content-Length, Accept-Ranges**, etc.
- **Response Body** – Contains the **actual data returned by the server**, such as HTML, CSS, JSON, or other resources (e.g., CSS code like `nav.navbar { ... }`).

Creating an Express Server

1. Create a project folder

```
mkdir express-server
cd express-server
```

2. Initialize a Node.js project

```
npm init -y
```

This creates a `package.json` file for the project.

3. Install Express

```
npm install express
```

4. Create the server file

Create a file named:

```
server.js
```

5. Import Express

```
const express = require("express");
```

6. Create an Express application

```
const app = express();
```

7. Create a route

```
app.get("/", (req, res) => {
  res.send("Hello from Express Server");
});
```

8. Start the server

```
app.listen(3000, () => {
  console.log("Server running on port 3000");
});
```

9. Run the server

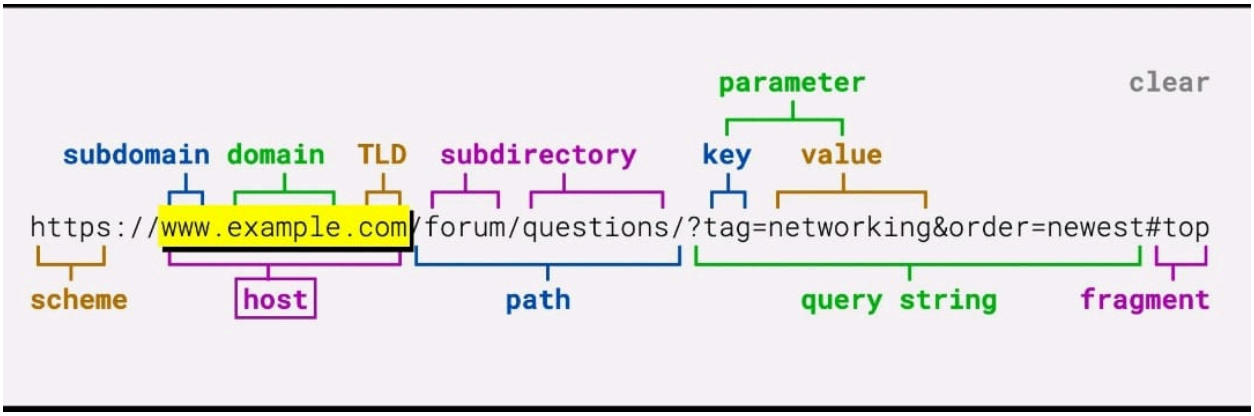
```
node server.js
```

Open in browser:

```
http://localhost:3000
```

You will see:

URL



1. **Scheme (Protocol)**

   `https://`

   - Defines the protocol used to communicate with the server.
   - Common protocols: **HTTP, HTTPS, FTP**.

2. **Subdomain**

   `www`

   - A subdivision of the main domain used to organize sections of a website.

3. **Domain**

   `example`

   - The main name of the website that identifies it on the internet.

4. **Top Level Domain (TLD)**

   `.com`

   - The last part of the domain name.
   - Examples: `.com`, `.org`, `.net`, `.edu`.

5. **Host**

   `www.example.com`

   - The complete domain name (subdomain + domain + TLD) that points to the server.

6. **Path**

   `/forum/questions`

   - Specifies the exact resource or page on the server.

7. **Query String**

   `?tag=networking&order=newest`

   - Contains parameters sent to the server to filter or modify the response.

8. **Parameters (Key–Value pairs)**

   - `tag = networking`
   - `order = newest`
   - These are used by the server to process the request dynamically.

9. **Fragment**

   `#top`

   - Points to a specific section inside the webpage.

Note:

The **fragment (`#...`) is never sent to the server**. It is handled entirely by the **browser** to navigate within the page, so the server has **no knowledge of it**.

Creating a TODO Application

Routes

- `POST /todo` → **Create a new todo**
- `GET /users` → **Get all users**

Features (CRUD)

- **Create Todo** → Add a new todo
- **View Todo** → Read existing todos
- **Edit Todo** → Update a todo
- **Delete Todo** → Remove a todo

These operations form **CRUD**:

- **C** → Create

- **R** → Read
- **U** → Update
- **D** → Delete

HTTP Methods Used

- **POST** → Create a new resource
- **GET** → Retrieve data
- **PUT** → Update an entire resource
- **PATCH** → Partially update a resource
- **DELETE** → Remove a resource

Other HTTP Methods

- **HEAD** → Same as GET but returns only headers (no body)
- **OPTIONS** → Shows allowed HTTP methods for a resource
- **CONNECT** → Establishes a network tunnel (used in proxies)
- **TRACE** → Returns the request for debugging purposes

In-Memory TODO API (Express)

1️⃣ Setup Express

```
npm init -y
npm install express
```

Create `server.js`

server.js

```
const express = require("express");
const app = express();

app.use(express.json());

let todos = [
  { id: 1, task: "Learn Express", completed: false },
  { id: 2, task: "Build REST API", completed: false }
];
```

2️⃣ GET → Read Todos

```
app.get("/todos", (req, res) => {
  res.json(todos);
});
```

Get single todo

```
app.get("/todos/:id", (req, res) => {
  const todo = todos.find(t => t.id === parseInt(req.params.i
d));
  res.json(todo);
});
```

3️⃣ POST → Create Todo

```
app.post("/todos", (req, res) => {
  const newTodo = {
    id: Date.now(),
    task: req.body.task,
    completed: false
  };
```

```
  todos.push(newTodo);
  res.status(201).json(newTodo);
});
```

Example request body

```
{
  "task": "Study REST APIs"
}
```

4️⃣ PUT → Update Entire Todo

```
app.put("/todos/:id", (req, res) => {
  const id = parseInt(req.params.id);

  const index = todos.findIndex(t => t.id === id);

  todos[index] = {
    id,
    task: req.body.task,
    completed: req.body.completed
  };

  res.json(todos[index]);
});
```

PUT replaces the **whole resource**.

5️⃣ PATCH → Partial Update

```
app.patch("/todos/:id", (req, res) => {
  const todo = todos.find(t => t.id === parseInt(req.params.i
```

```
d));

  if (req.body.task !== undefined) {
    todo.task = req.body.task;
  }

  if (req.body.completed !== undefined) {
    todo.completed = req.body.completed;
  }

  res.json(todo);
});
```

PATCH updates **only specific fields**.

6️⃣ DELETE → Remove Todo

```
app.delete("/todos/:id", (req, res) => {
  const id = parseInt(req.params.id);

  todos = todos.filter(t => t.id !== id);

  res.json({ message: "Todo deleted" });
});
```

7️⃣ HEAD Method

Returns **headers only (no body)**.

```
app.head("/todos", (req, res) => {
  res.status(200).end();
});
```

8️⃣ OPTIONS Method

Shows **allowed HTTP methods**.

```
app.options("/todos", (req, res) => {
  res.set("Allow", "GET, POST, PUT, PATCH, DELETE, HEAD, OPTI
ONS");
  res.send();
});
```

9️⃣ TRACE Method (rarely used)

```
app.trace("/trace", (req, res) => {
  res.send(req.headers);
});
```

🔟 Start Server

```
app.listen(3000, () => {
  console.log("Server running on port 3000");
});
```

Example API Endpoints

```
GET     /todos
GET     /todos/:id
POST    /todos
PUT     /todos/:id
PATCH   /todos/:id
DELETE  /todos/:id
HEAD    /todos
```

```
OPTIONS /todos
TRACE   /trace
```

Headers

Request Headers

Header	Use Case	Example
Host	Specifies the target server/domain where the request is sent	`Host: www.example.com`
Origin	Indicates the origin of the request (used in CORS validation)	`Origin: https://example.com`
Referer	Shows the previous webpage from which the request was made	`Referer: https://example.com/previous-page`
User-Agent	Identifies the client making the request (browser, OS, device)	`User-Agent: Mozilla/5.0 Chrome/119`
Accept	Specifies the response content type the client expects	`Accept: application/json`
Accept-Language	Indicates the preferred language for the response	`Accept-Language: en-US,en;q=0.9`
Accept-Encoding	Lists the compression formats supported by the client	`Accept-Encoding: gzip, deflate, br`
Connection	Controls whether the TCP connection stays open or closes	`Connection: keep-alive`
Authorization	Sends authentication credentials (token, API key, etc.)	`Authorization: Bearer <token>`
Cookie	Sends stored cookies from the browser to the server	`Cookie: sessionId=abc123`
If-Modified-Since	Used for cache validation; server returns data only if modified after the given date	`If-Modified-Since: Wed, 21 Oct 2015 07:28:00 GMT`

Header	Use Case	Example
Cache-Control	Defines caching behavior for the request	`Cache-Control: no-cache`

Response Headers

Header	Use Case	Example
Content-Type	Specifies the format of the response body	`Content-Type: application/json`
Content-Length	Indicates the size of the response body in bytes	`Content-Length: 348`
Date	The date and time when the response was generated	`Date: Tue, 15 Nov 2026 08:12:31 GMT`
Server	Identifies the software used by the server	`Server: nginx/1.24.0`
Set-Cookie	Sends cookies from the server to be stored by the client	`Set-Cookie: sessionId=abc123; HttpOnly`
Cache-Control	Defines caching rules for the response	`Cache-Control: max-age=3600`
ETag	A unique identifier for a specific version of a resource (used for caching)	`ETag: "abc123xyz"`
Last-Modified	Shows the last time the resource was modified	`Last-Modified: Wed, 21 Oct 2015 07:28:00 GMT`
Location	Redirects the client to another URL	`Location: https://example.com/new-page`
Access-Control-Allow-Origin	Defines which origins are allowed to access the resource (CORS)	`Access-Control-Allow-Origin: *`
Content-Encoding	Indicates the compression used in the response	`Content-Encoding: gzip`
Connection	Controls whether the network connection stays open or closes	`Connection: keep-alive`

HTTP Status Codes

1xx — Informational

Code	Meaning	Description
100	Continue	Request received, client should continue sending body
101	Switching Protocols	Server switching protocols (e.g., HTTP → WebSocket)
102	Processing	Server has received request and is processing

2xx — Success

Code	Meaning	Description
200	OK	Request succeeded
201	Created	Resource successfully created
202	Accepted	Request accepted but processing not complete
204	No Content	Success but no response body

3xx — Redirection

Code	Meaning	Description
301	Moved Permanently	Resource permanently moved
302	Found	Temporary redirect
303	See Other	Redirect to another resource
304	Not Modified	Resource not modified (cache)
307	Temporary Redirect	Same request method but temporary redirect
308	Permanent Redirect	Permanent redirect with same method

4xx — Client Errors

Code	Meaning	Description
400	Bad Request	Invalid request syntax
401	Unauthorized	Authentication required

Code	Meaning	Description
403	Forbidden	Server refuses request
404	Not Found	Resource not found
405	Method Not Allowed	HTTP method not supported
408	Request Timeout	Server timed out waiting
409	Conflict	Request conflicts with current state
413	Payload Too Large	Request body too large
415	Unsupported Media Type	Unsupported content type
429	Too Many Requests	Rate limit exceeded

5xx — Server Errors

Code	Meaning	Description
500	Internal Server Error	Generic server failure
501	Not Implemented	Server doesn't support functionality
502	Bad Gateway	Invalid response from upstream server
503	Service Unavailable	Server temporarily unavailable
504	Gateway Timeout	Upstream server timeout